Eugenio Pasos

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SUMMARY

Robotics and Systems Engineer with experience in software development, avionics, and hardware integration. Proficient in ROS2, Python, and C/C++. Focused on implementing software and hardware solutions to optimize robotic performance

EDUCATION

M.S. in Electrical Engineering Expected: May 2027
San Jose State University – San Jose, CA | GPA: 3.82

B.S. in Electrical Engineering June 2025
California Polytechnic State University – San Luis Obispo, CA
Cum Laude | GPA: 3.55 | Dean's List – 7 Quarters | President's List – 2024, 2025

Relevant Coursework:
Autonomous Mobile Robots, Mechatronics, Vision-Based Robotic Manipulation, Neural Networks, Computer Vision, Digital Control Systems, Microcontroller-Based Systems Design, Classical Control Systems, Digital Signal Processing, Continuous-Time Signals & Systems, Discrete-Time Systems & Signals, Semiconductor Electronics

WORK EXPERIENCE

ROS System Engineer Intern – Renesas Electronics June 2026 – Present
- Assembled and integrated an Amazing Hand 8-DoF robotic hand; authored custom hardware control drivers and integrated MediaPipe hand detection with ros2_control to achieve real-time hand-mimic tracking
- Engineered a corresponding Foxglove simulation environment for the Amazing Hand to enable real-time telemetry and diagnostic visualization on the Renesas Robotics Development Kit

Associate Avionics Engineer – Astranis Space Technologies January 2026 – May 2026
- Completed design and layout for two GaNFET load switch boards in Altium to model flight-like power switching
- Wrote OpenHTF Python code for hardware validation for shift registers, pressure transducer amplifiers, and valve actuation
- Led cross-functional schematic review for next-generation flight board, coordinating with thermal, software, harnessing, and avionics teams to ensure proper board design

PROJECT EXPERIENCE

Autonomous Mapping Rover – Senior Project January 2025 – June 2025
- Designed and built an autonomous mapping rover for indoor environments in Python and C++ using ROS2
- Integrated LiDAR, IMU, and encoder data for real-time navigation, path planning, and obstacle avoidance
- Implemented and tested 2D SLAM mapping using slam_toolbox and autonomous navigation with Nav2

Autonomous Pick and Place Robot – Robotic Vision September 2024 – December 2024
- Developed Python code to control a robot arm to sort balls into different bins based on color autonomously
- Utilized OpenCV to detect red, blue, yellow, and orange colored spheres using an RGB camera
- Derived forward and inverse kinematics for the OpenManipulator-X 4-revolute joint robot arm

Mock Driving Simulator – Embedded Systems April 2024 – June 2024
- Developed a C-based driving simulator using two STM32 boards to manage the controller and infotainment systems
- Wrote C code to control a DC motor via PWM, handling SPI, UART, & CAN communication for IO peripherals

LEADERSHIP EXPERIENCE

Low Voltage Team Lead – Prototype Vehicles Laboratory January 2024 – June 2025
Low Voltage Team Member – Prototype Vehicles Laboratory January 2023 – December 2023
Eagle Scout – Boy Scouts of America December 2021

SKILLS

Programming Languages: Python, C, C++, MATLAB, SystemVerilog
Software: ROS2, Linux, Docker, GIT, Matlab, LTSpice, Altium, Vivado, Jira
Test Equipment Proficiency: Oscilloscope, Function Generator, Power Supply, Sourcemeter, DMM